

2. Write and execute SQL Data Definition Language (DDL) commands such as CREATE, ALTER, DROP, RENAME, and TRUNCATE to define and modify tables. Insert data into the tables and apply appropriate integrity constraints such as NOT NULL, UNIQUE, PRIMARYKEY, FOREIGN KEY, and CHECK.

```
mysql> create database Dept;
```

```
Query OK, 1 row affected (0.77 sec)
```

```
mysql> use Dept;
```

```
Database changed
```

Department Table

```
mysql> CREATE TABLE Department (Dept_ID INT PRIMARY KEY,Dept_Name  
VARCHAR(50) UNIQUE NOT NULL);
```

```
Query OK, 0 rows affected (0.09 sec)
```

Student Table

```
mysql> CREATE TABLE Student (Stud_ID INT PRIMARY KEY,Stud_Name VARCHAR(50)  
NOT NULL,Age INT CHECK (Age >= 18),Email VARCHAR(100) UNIQUE,Dept_ID  
INT,FOREIGN KEY (Dept_ID) REFERENCES Department(Dept_ID));
```

```
Query OK, 0 rows affected (0.06 sec)
```

INSERT Data into Tables

```
mysql> INSERT INTO Department VALUES(1, 'Computer Engineering'),(2, 'Information  
Technology'),(3, 'Electronics');
```

```
Query OK, 3 rows affected (0.02 sec)
```

```
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from Department;
```

```

+-----+-----+
| Dept_ID | Dept_Name      |
+-----+-----+
|    1    | Computer Engineering |
|    3    | Electronics         |
|    2    | Information Technology |
+-----+-----+

```

3 rows in set (0.01 sec)

```
mysql> INSERT INTO Student VALUES(101, 'Amit Sharma', 20, 'amit@gmail.com', 1),(102,
'Neha Patil', 19, 'neha@gmail.com', 2),(103, 'Rahul Verma', 21, 'rahul@gmail.com', 1);
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> select * from Student;
```

```

+-----+-----+-----+-----+-----+
| Stud_ID | Stud_Name  | Age | Email          | Dept_ID |
+-----+-----+-----+-----+-----+
|   101   | Amit Sharma |  20 | amit@gmail.com |    1    |
|   102   | Neha Patil  |  19 | neha@gmail.com  |    2    |
|   103   | Rahul Verma |  21 | rahul@gmail.com |    1    |
+-----+-----+-----+-----+-----+

```

3 rows in set (0.00 sec)

ALTER TABLE

```
mysql> ALTER TABLE Student ADD City VARCHAR(30);
```

Query OK, 0 rows affected (0.03 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> select * from Student;
```

```

+-----+-----+-----+-----+-----+-----+
| Stud_ID | Stud_Name | Age | Email | Dept_ID | City |
+-----+-----+-----+-----+-----+-----+
| 101 | Amit Sharma | 20 | amit@gmail.com | 1 | NULL |
| 102 | Neha Patil | 19 | neha@gmail.com | 2 | NULL |
| 103 | Rahul Verma | 21 | rahul@gmail.com | 1 | NULL |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

RENAME TABLE

```
mysql> RENAME TABLE Student TO Student_Details;
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> select * from Student_Details;
```

```

+-----+-----+-----+-----+-----+-----+
| Stud_ID | Stud_Name | Age | Email | Dept_ID | City |
+-----+-----+-----+-----+-----+-----+
| 101 | Amit Sharma | 20 | amit@gmail.com | 1 | NULL |
| 102 | Neha Patil | 19 | neha@gmail.com | 2 | NULL |
| 103 | Rahul Verma | 21 | rahul@gmail.com | 1 | NULL |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

```

```
mysql> show tables;
```

```

+-----+
| Tables_in_dept |
+-----+
| department |
| student_details |
+-----+

```

2 rows in set (0.02 sec)

TRUNCATE TABLE

```
mysql> TRUNCATE TABLE Student_Details;
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> select * from Student_Details;
```

Empty set (0.00 sec)

```
mysql> desc Student_Details;
```

Field	Type	Null	Key	Default	Extra
Stud_ID	int	NO	PRI	NULL	
Stud_Name	varchar(50)	NO		NULL	
Age	int	YES		NULL	
Email	varchar(100)	YES	UNI	NULL	
Dept_ID	int	YES	MUL	NULL	
City	varchar(30)	YES		NULL	

6 rows in set (0.01 sec)

DROP TABLE

```
mysql> DROP TABLE Student_Details;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> select * from Student_Details;
```

ERROR 1146 (42S02): Table 'dept.student_details' doesn't exist

```
mysql> show tables;
```

```

+-----+
| Tables_in_dept |
+-----+
| department     |
+-----+
1 row in set (0.00 sec)

```

Applying Integrity Constraints in SQL

```
mysql> CREATE TABLE Course (Course_ID INT PRIMARY KEY, Course_Name
VARCHAR(50) NOT NULL UNIQUE, Duration INT CHECK (Duration >= 1));
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> desc Course;
```

```

+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Course_ID  | int       | NO   | PRI | NULL    |       |
| Course_Name | varchar(50) | NO   | UNI | NULL    |       |
| Duration   | int       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+

```

```
mysql> show tables;
```

```

+-----+
| Tables_in_dept |
+-----+
| course         |
| department     |
+-----+
2 rows in set (0.00 sec)

```

```
mysql> CREATE TABLE Student (Student_ID INT PRIMARY KEY,Student_Name
VARCHAR(50) NOT NULL,Email VARCHAR(100) UNIQUE,Age INT CHECK (Age >=
18),Course_ID INT,FOREIGN KEY (Course_ID) REFERENCES Course(Course_ID));
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> show tables;
```

```
+-----+
| Tables_in_dept |
+-----+
| course      |
| department  |
| student     |
+-----+
```

3 rows in set (0.00 sec)

```
mysql> desc Student;
```

```
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| Student_ID | int       | NO   | PRI | NULL    |      |
| Student_Name | varchar(50) | NO   |     | NULL    |      |
| Email       | varchar(100) | YES  | UNI | NULL    |      |
| Age        | int       | YES  |     | NULL    |      |
| Course_ID  | int       | YES  | MUL | NULL    |      |
+-----+-----+-----+-----+-----+-----+
```

5 rows in set (0.00 sec)